

Finite-Jet Pressure Laws for Gevrey Reversion and Condensation

Complete multiplicative asymptotics,
explicit critical constants, and Poisson centering

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Abstract

The ProveIt research corpus leaves a precise gap in its sharp fixed-coefficient-ball analysis of formal reversion: at Gevrey orders $0 < s < 1/2$, the leading logarithmic amplification is known, but a multiplicative equivalent is not supplied. We close that gap for the stated factorial weights and, more generally, for positive weights with an eventually shifted-factorial tail and an arbitrary finite head. The logarithm of the normalized extremal inverse coefficient is a finite sum $\sum_{ks \leq 1} p_k n^{1-ks} + o(1)$, with an explicit coefficient extraction formula for every p_k . Thus every positive order, including all reciprocal-integer transition points, has a complete multiplicative equivalent.

A parameter $\theta \geq -1$ puts the reversion extremum ($\theta = 1$) and superexponentially weighted plane trees ($\theta = -1$) into one theorem. The proof reduces an exact Lagrange sum to finitely many small-part sizes and evaluates it by a Poisson change of measure. It gives a finite-head universality law, all critical windows $s = 1/r + t/\log n$, and explicit all-order centering parameters for the asymptotically independent small-part counts. Exact algebraic checks and non-certified numerical diagnostics accompany the proofs.

Status. Conventional mathematical proofs, not Lean-verified or peer-reviewed. The identified repository gap is resolved under the hypotheses stated here; no claim of worldwide publication priority or resolution of an unrelated longstanding conjecture is made.

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1 The research target and its boundaries

1.1 A specific missing asymptotic, not another type theorem

For a formal power series $f(z) = z + O(z^2)$, let $f^{(-1)}$ denote its compositional inverse. Consider the fixed coefficient ball

$$\mathcal{B}_{s,A,C} = \left\{ z + \sum_{n \geq 2} f_n z^n : |f_n| \leq C A^{n-1} ((n-1)!)^s \right\}, \quad s, A, C > 0. \quad (1.1)$$

Preservation of an n th-root coefficient type under inversion does not specify the amplification at these exact constants A and C . Subexponential factors disappear under n th roots, but can still become arbitrarily large.

The inspected ProveIt package *Sharp Subexponential Reversion* [2] identifies a single extremizer for (1.1) and proves its sharp leading logarithm. It gives multiplicative equivalents above order $1/2$ and at $1/2$, but explicitly leaves the corresponding answer below $1/2$ open within that article. This is the target of the present work. The earlier exact weighted-type theorem [3] is not claimed as a new contribution here.

The repository snapshot used for the inspection is

a4268e78 ebd0f6bf 71ba609c 07f4b341 5748f532.

The relevant directories lie under

Analysis/Transseries/docs/series-and-transseries/.

The package names in the bibliography, source headings, and this snapshot are more durable locators than mutable source line numbers. The source review was targeted: the relevant READMEs, the predecessor's extremal and localization arguments, and searches for overlapping reversion results were inspected. This was not an audit of the entire repository.

1.2 What is proved here

The principal theorem supplies, for every fixed $s > 0$, a finite list of explicit constants that determines the extremal inverse coefficient up to a relative error tending to zero. For example, the previously uncovered order $s = 1/3$ has the formula

$$\begin{aligned} \log R_n = C n^{2/3} + \left(2^{1/3} C + \frac{2}{3} C^2 \right) n^{1/3} \\ + 6^{1/3} C + \frac{5}{3} 2^{1/3} C^2 + \frac{7}{9} C^3 + o(1), \end{aligned} \quad (1.2)$$

where R_n is the supremum of the normalized inverse coefficient over (1.1). The formula has an actual constant term: omitting it would give an incorrect multiplicative equivalent.

There are three additional consequences. First, only the initial $\lfloor 1/s \rfloor$ weights affect the normalized multiplicative asymptotic. Second, every reciprocal integer $1/r$ has an explicitly resolved $1/\log n$ transition window. Third, the same finite implicit equation computes all centering corrections needed for Gaussian limits of the accompanying small-part counts. A one-parameter family makes these statements apply to both extremal reversion and weighted trees.

1.3 Published precedents and novelty qualifications

Formal Lagrange inversion is classical; we use its coefficient form as presented, for example, by Gessel [4]. Factorially divergent coefficient algebras and their composition and inversion rules have substantial antecedents, including Borinsky [6].

For superexponential tree weights, Janson, Jonsson and Stefánsson [5] already prove condensation, Gaussian and critical Poisson limits, and total-variation approximation by independent Poisson variables. Their Remark 2.8 gives the first two exponent terms and points toward higher corrections. Those probabilistic phenomena are not discoveries of this article. Our contribution is the explicit all-order pressure and centering formula, its reversion application, and the finite-head and transition-window consequences proved below.

The principal proofs are self-contained after formal Lagrange inversion and elementary facts about factorials and Poisson variables. In particular, no unreviewed asymptotic theorem from the repository is needed as a black-box input. The literature check is not an exhaustive priority search; the manuscript should receive independent mathematical review before its conclusions are treated as established results.

Statement	Scope and status
Exact extremizer	Known in the predecessor; short proof included.
All-order exponent coefficients	Explicit formula and proof for every fixed $s > 0$.
Complete multiplicative equivalent	Proved for eventually shifted-factorial tails and arbitrary positive finite heads.
All reciprocal critical windows	Proved, uniformly on bounded window parameters.
Poisson small-part law	Generalized and given explicit centers; the tree phenomenon has published precedents.
Numerical evidence	Finite exact algebra checks plus floating-point diagnostics; not an asymptotic proof.
Beyond the theorem's hypotheses	General oscillatory tails, cancellation, and the joint limit $s \downarrow 0$ remain separate questions.

Table 1: Claim and dependency ledger.

2 The main theorem

2.1 Weights and the interpolation family

Definition 2.1 (Eventually shifted-factorial weights). Fix $s > 0$. A positive sequence $(w_j)_{j \geq 1}$ is admissible here if there exist $D_* > 0$, $\nu \geq 0$, and an integer $j_0 \geq 1$ such that

$$w_j = D_* \Gamma(j + \nu + 1)^s \quad (j \geq j_0). \quad (2.1)$$

The finitely many weights $w_1, \dots, w_{j_0-1}$ may be arbitrary positive numbers. In particular, log-convexity of the entire sequence is not required.

The weights in the predecessor are the special cases

$$w_j = \left(\frac{\Gamma(j + \nu + 1)}{\Gamma(\nu + 2)} \right)^s, \quad \nu \in \{0, 1\}. \quad (2.2)$$

They satisfy $w_1 = 1$. The shift $\nu = 0$ gives $w_j = (j!)^s$; $\nu = 1$ gives $w_j = ((j + 1)!/2)^s$.

Fix $C > 0$ and $\theta \geq -1$. All the following definitions involving an infinite series are formal. Put

$$W(z) = \sum_{j \geq 1} w_j z^j, \quad F_\theta(z) = \begin{cases} z(1 - \theta C W(z))^{1/\theta}, & \theta \neq 0, \\ z \exp(-C W(z)), & \theta = 0, \end{cases} \quad (2.3)$$

and define

$$G_\theta = F_\theta^{(-1)}, \quad R_n^{(\theta)} = \frac{[z^n]G_\theta(z)}{Cw_{n-1}}, \quad n \geq 2. \quad (2.4)$$

The linear coefficient of F_θ is 1, so the inverse exists uniquely in $\mathbb{R}[[z]]$. We shall prove that $R_n^{(\theta)} > 0$. The restriction $\theta \geq -1$ is a positivity condition on finite Lagrange coefficients, not an analyticity assumption on W .

At $\theta = 1$, $F_1 = z - C \sum_{j \geq 1} w_j z^{j+1}$ is the extremal negative series. At $\theta = -1$, G_{-1} satisfies

$$G_{-1}(z) = z(1 + CW(G_{-1}(z))), \quad (2.5)$$

the formal equation for plane rooted trees with leaf weight 1 and weight Cw_j for a vertex of outdegree $j \geq 1$.

2.2 A finite polynomial determines the answer

Choose an integer $J \geq 1$ such that

$$(J+1)s > 1. \quad (2.6)$$

Define the finite polynomial and its rational transform

$$a_j = Cw_j, \quad A(t) = \sum_{j=1}^J a_j t^j, \quad B(t) = \frac{tA'(t)}{1 - \theta A(t)}. \quad (2.7)$$

Unlike W , these expressions are analytic near the origin. For any real a , let

$$\Psi_a(u) = \int_0^u (1-v)^{-a} dv = \sum_{r \geq 1} \frac{(a)_{r-1}}{r!} u^r, \quad (2.8)$$

where the rising factorial of length zero is 1. Thus $\Psi_1(u) = -\log(1-u)$; for $a \neq 1$, $\Psi_a(u) = ((1-u)^{1-a} - 1)/(a-1)$.

Theorem 2.2 (Finite-jet pressure law). *Under [definition 2.1](#), fix $C > 0$, $\theta \geq -1$, and J satisfying [\(2.6\)](#). Define*

$$p_k = \frac{1}{k} [t^k] \Psi_{ks}(B(t)), \quad 1 \leq k \leq J. \quad (2.9)$$

Then

$$\boxed{\log R_n^{(\theta)} = \sum_{k=1}^J p_k n^{1-ks} + o(1).} \quad (2.10)$$

Equivalently,

$$R_n^{(\theta)} = \exp \left\{ \sum_{k=1}^J p_k n^{1-ks} \right\} (1 + o(1)). \quad (2.11)$$

For a fixed s , terms with $ks > 1$ may be omitted. The coefficient p_k depends only on $s, \theta, a_1, \dots, a_k$, not on the rest of the weight sequence or the chosen truncation order $J \geq k$.

In particular, when $0 < s \leq 1$, taking $J = \lfloor 1/s \rfloor$ supplies exactly the terms that do not tend to zero. For $s > 1$, taking $J = 1$ shows $R_n^{(\theta)} \rightarrow 1$. At $s = 1$ the limit is $\exp(Cw_1)$, independent of θ .

Here and throughout, a complete multiplicative equivalent means a formula with relative error $o(1)$. It does not mean that every vanishing correction after that relative error has been calculated.

Theorem 2.3 (Finite analytic pressure). *With the same hypotheses, let $y = y(x)$ be the unique analytic germ with $y(0) = 0$ satisfying*

$$y = x(1 - B(y))^{-s}. \quad (2.12)$$

Set $d(x) = B(y(x))$ and

$$P(x) = L_\theta(A(y(x))) + s(\log(1 - d(x)) + d(x)), \quad (2.13)$$

where

$$L_\theta(v) = \begin{cases} -\theta^{-1} \log(1 - \theta v), & \theta \neq 0, \\ v, & \theta = 0. \end{cases} \quad (2.14)$$

For sufficiently large n , this real germ is defined at $x = n^{-s}$, and

$$R_n^{(\theta)} = \exp\{nP(n^{-s})\}(1 + o(1)), \quad xP'(x) = d(x). \quad (2.15)$$

Its Taylor coefficients through order J are precisely (2.9).

The pressure in (2.13) uses a *finite* polynomial. There is no substitution of a positive real number into the divergent series W . This distinction is necessary: replacing A by CW in an analytic saddle equation would not be justified.

2.3 Uniformity needed later

The conclusions of theorems 2.2 and 2.3 hold uniformly when s, C , and θ vary in compact subsets of $(0, \infty)$, $(0, \infty)$, and $[-1, \infty)$, respectively, provided one fixed J has $(J + 1)s_{\min} > 1$ and the tail description is uniform. For example, one may fix j_0 and ν , let D_* and the finitely many exceptional weights be positive continuous functions of the parameters, and bound them above and away from zero on those compact sets. This version includes (2.2) with varying s . The proof below identifies the uniform estimates explicitly.

3 Exact Lagrange coefficients and the extremum

3.1 The positive finite sum

Let $m = n - 1$, and write

$$P_w(m, k) = \sum_{\substack{j_1 + \dots + j_k = m \\ j_i \geq 1}} \prod_{i=1}^k w_{j_i}. \quad (3.1)$$

This is a finite sum over ordered positive compositions of m .

Proposition 3.1 (Exact coefficient formula). *For $n \geq 2$,*

$$R_n^{(\theta)} = \frac{1}{nCw_m} \sum_{k=1}^m \frac{\prod_{r=0}^{k-1} (n + \theta r)}{k!} C^k P_w(m, k). \quad (3.2)$$

Every summand is positive and the $k = 1$ summand equals 1.

Proof. Formal Lagrange inversion [4] gives

$$[z^n]G_\theta = \frac{1}{n} [t^{n-1}] (1 - \theta CW(t))^{-n/\theta}$$

for $\theta \neq 0$, with $\exp(nCW(t))$ at $\theta = 0$. The coefficient of $(CW)^k$ is $\prod_{r=0}^{k-1} (n + \theta r)/k!$, including the continuous $\theta = 0$ interpretation. Since $k \leq n - 1$ and $\theta \geq -1$, all factors are strictly positive. Expansion of W^k gives (3.2); its first term is 1 after normalization. $\square$

No convergence is used in this argument. A coefficient of degree m in a power of W depends on only $w_1, \dots, w_m$.

Proposition 3.2 (Simultaneous extremality). *For arbitrary positive weights and $A_0, C > 0$, let*

$$\mathcal{B}(A_0, C; w) = \left\{ z + \sum_{j \geq 1} f_{j+1} z^{j+1} : |f_{j+1}| \leq C A_0^j w_j \right\}.$$

The single series

$$F_*(z) = z - C \sum_{j \geq 1} A_0^j w_j z^{j+1} \quad (3.3)$$

attains

$$\sup_{f \in \mathcal{B}(A_0, C; w)} \frac{|[z^n] f^{(-1)}|}{C A_0^{n-1} w_{n-1}} = R_n^{(1)} \quad (3.4)$$

for every $n \geq 2$ simultaneously.

Proof. For $f = z(1+H)$, Lagrange inversion expresses the inverse coefficient as $\frac{1}{n}[z^{n-1}](1+H)^{-n}$. The coefficients of $(1-u)^{-n}$ are nonnegative. Taking absolute values and majorizing each coefficient of H by $C A_0^j w_j$ therefore gives the coefficient of $F_*^{(-1)}$. For F_* every contribution has the same positive sign, so equality holds. Finally, replacing z by $A_0 z$ contributes the common factor A_0^{n-1} . This proves both the extremum and its independence of A_0 after normalization. $\square$

This proposition is an exact statement about a coefficient ball, not about the inverse of an arbitrary positive or signed series. The asymptotic conclusions about this supremum do not assert that every member of the ball attains it asymptotically.

4 Localization at one large part

The first reduction is from all compositions to those with a unique part near $m = n - 1$. The argument also records why a fixed finite head of the weights does not obstruct the analysis.

Lemma 4.1 (Convolution bound). *Under [definition 2.1](#), there is a constant $K \geq 1$ such that*

$$\sum_{j=1}^{m-1} w_j w_{m-j} \leq K w_{m-1} \quad (m \geq 2), \quad P_w(m, k) \leq K^{k-1} w_{m-k+1}. \quad (4.1)$$

Proof. There are constants $c_1, c_2 > 0$ such that, for every $j \geq 1$,

$$c_1 (j!)^s (j+1)^{s\nu} \leq w_j \leq c_2 (j!)^s (j+1)^{s\nu}. \quad (4.2)$$

This follows from the gamma quotient asymptotic and adjustment of finitely many constants; for integer ν it follows directly from a finite product.

For large m , and $1 \leq j \leq m/2$, both $m-j$ and $m-1$ are in the exact tail. Their quotient is

$$\frac{w_{m-j}}{w_{m-1}} = \left(\prod_{r=1}^{j-1} (m-j+\nu+r) \right)^{-s} \leq j^{-s(j-1)}.$$

Consequently

$$\frac{w_j w_{m-j}}{w_{m-1}} \leq c_2 (j!)^s (j+1)^{s\nu} j^{-s(j-1)}.$$

Stirling's upper bound makes the right side at most a constant times $j^{s\nu+3s/2}e^{-sj}$. This is summable. Reflection about $m/2$ and an increase of K for the remaining finitely many m prove the first assertion.

For the second, induct on k . Splitting off the first part and using the induction hypothesis gives

$$P_w(m, k) \leq K^{k-2} \sum_{j=1}^{m-k+1} w_j w_{m-k+2-j} \leq K^{k-1} w_{m-k+1}.$$

The case $k = 1$ is an equality. □

Let $T_{n,k}$ be the k th summand in (3.2). The elementary estimate $\prod_{r=0}^{k-1} (n + \theta r) \leq (c_\theta n)^k$, where $c_\theta = 1 + \max(\theta, 0)$, combines with lemma 4.1 to give constants $c, D > 0$ such that

$$T_{n,k} \leq c \frac{(Dn^{1-s})^{k-1}}{k!} \quad (1 \leq k \leq n/2) \quad (4.3)$$

for large n . Indeed, the quotient w_{n-k}/w_{n-1} consists of $k-1$ inverse s th powers of factors at least $n/2$. For $k > n/2$, monotonicity in the exact tail, with a fixed adjustment for the finite head, gives

$$\sum_{k > n/2} T_{n,k} \leq \exp\{-\tfrac{s}{2}n \log n + O(n)\}. \quad (4.4)$$

To check the entropy factor in this last bound, note that $(c_\theta n)^k/k!$ is $\exp(O(n))$ when $k > n/2$; $C^{k-1}K^{k-1}$ has the same bound. The remaining weight quotient has logarithm at most $-\tfrac{s}{2}n \log n + O(n)$.

Lemma 4.2 (One-large-part localization). *Choose real exponents satisfying*

$$\max(1-s, 0) < \kappa < \alpha < 1, \quad h = \lfloor n^\alpha \rfloor. \quad (4.5)$$

For large n , the total contribution to (3.2) from compositions with no part at least $m-h$ is $o(1)$. More precisely, it is bounded by a sum of terms of the forms $\exp(-cn^\alpha \log n)$ and $\exp(-c'n^\kappa \log n)$, with positive constants.

Proof. First restrict the number of parts to $k \leq t = \lfloor n^\kappa \rfloor$. Since $h < m/2$ eventually, concentration of a factorial product under an upper bound $m-h$ gives

$$\prod_{i=1}^k j_i! \leq (m-h)!h!. \quad (4.6)$$

One proof permits zero parts and moves a unit from a smaller positive part to a larger part that is below the cap. Such a move does not decrease the factorial product. At a maximum at most two parts are nonzero, and they have sizes $m-h$ and h .

By (4.2), the normalized weight product is at most

$$c_1^{-1} c_2^k \binom{m}{h}^{-s} \prod_{i=1}^k (j_i + 1)^{s\nu} \leq c_1^{-1} c_2^k \binom{m}{h}^{-s} (1 + m/k)^{s\nu k}.$$

We have discarded a denominator $(m+1)^{s\nu} \geq 1$. There are $\binom{m-1}{k-1}$ positive compositions of length k . The logarithms of this count, of the Lagrange factor, and of all fixed constants together are $O(k \log(en/k) + \log n)$. The total for $k \leq t$ is thus bounded by

$$\exp \left\{ -s \log \binom{m}{h} + O \left(t \log \frac{en}{t} + \log n \right) \right\}. \quad (4.7)$$

Here $\log \binom{m}{h} = (1 - \alpha + o(1))n^\alpha \log n$, which dominates the positive error because $\alpha > \kappa$.

For $k > t$, use (4.3) and (4.4). The ratio t/n^{1-s} tends to infinity. The inequality $k! \geq (k/e)^k$, followed by a geometric tail estimate, gives

$$\sum_{t < k \leq n/2} T_{n,k} \leq \exp \left\{ -t \log \frac{t}{eDn^{1-s}} + O(\log n) \right\}.$$

Its exponent is a negative constant multiple of $n^\kappa \log n$. These bounds prove the claim. $\square$

The error in this lemma is *absolute* in the normalized quantity $R_n^{(\theta)}$. Since $R_n^{(\theta)} \geq 1$, an absolute $o(1)$ is also harmless for a multiplicative equivalent. In the next reduction, where sums can be exponentially large on a sublinear scale, a relative estimate will be needed instead.

5 The exact cloud and the finite-size cutoff

5.1 Grouping the small parts

In a localized composition the part of size at least $m - h$ is unique. Call it the large part and write its size as $m - d$. If m_j counts the remaining parts of size j , then

$$\ell = \sum_{j \geq 1} m_j, \quad d = \sum_{j \geq 1} j m_j \leq h. \quad (5.1)$$

There are $(\ell + 1)! / \prod_j m_j!$ ordered compositions with this multiplicity vector. Multiplication by the $k = \ell + 1$ Lagrange factor in (3.2) cancels both that factorial and the first factor n . We obtain the exact formula

$$R_{n,\text{loc}}^{(\theta)}(h) = \sum_{\substack{m_1, m_2, \dots \geq 0 \\ \sum j m_j \leq h}} \frac{\prod_{r=1}^{\ell} (n + \theta r)}{\prod_j m_j!} \frac{w_{m-d}}{w_m} \prod_{j \geq 1} a_j^{m_j}. \quad (5.2)$$

The all-zero multiplicity vector contributes 1. In this section $a_j = Cw_j$ is defined for all $j \geq 1$, not only $j \leq J$.

Lemma 5.1 (Only finitely many small sizes matter). *Let $S_{n,J}(h)$ denote the sum in (5.2) restricted to $m_j = 0$ for all $j > J$. Under (2.6) and (4.5),*

$$R_n^{(\theta)} = S_{n,J}(h)(1 + o(1)). \quad (5.3)$$

Proof. All summands are nonnegative. Compare a summand with one obtained by adding a part of size j while retaining $d + j \leq h$. The ratio is at most

$$\frac{\lambda_j^+}{m_j + 1}, \quad \lambda_j^+ = (n + \theta_+ h) a_j (n + \nu - h)^{-sj}, \quad \theta_+ = \max(\theta, 0). \quad (5.4)$$

For the first factor use $n + \theta(\ell + 1) \leq n + \theta_+ h$. For the large-part quotient, all relevant indices are eventually in the exact tail and each of the j denominator factors is at least $n + \nu - h$. This proves (5.4).

For j in the exact tail and $j < h$, successive intensities have ratio

$$\frac{\lambda_{j+1}^+}{\lambda_j^+} = \left(\frac{j + \nu + 1}{n + \nu - h} \right)^s = o(1)$$

uniformly in that range. At the finitely many exceptional indices the same conclusion follows from a fixed weight quotient times $O(n^{-s})$. Hence

$$\sum_{j=J+1}^h \lambda_j^+ = O(n^{1-(J+1)s}) = o(1). \quad (5.5)$$

Delete every part of size greater than J from a cloud. Its remaining low-size cloud is still admissible. Reinsert the deleted parts in a fixed order and apply (5.4) repeatedly. Summing their multiplicities bounds the ratio to the original low-size summand by $\prod_{j>J} \sum_{r \geq 0} (\lambda_j^+)^r / r!$. Extending the sum beyond the original size cutoff only increases it. Thus

$$S_{n,J}(h) \leq R_{n,\text{loc}}^{(\theta)}(h) \leq S_{n,J}(h) \exp \left(\sum_{j>J} \lambda_j^+ \right) = S_{n,J}(h)(1 + o(1)).$$

Finally, lemma 4.2 adds only $o(1)$, and $S_{n,J}(h) \geq 1$. This proves (5.3). $\square$

This is the source of finite-jet universality. A size j has a natural intensity of order n^{1-js} . Once this tends to zero, the total effect of all such omitted sizes is multiplicatively negligible. The proof justifies this intuition without evaluating the divergent weight generating function.

6 The interaction and its Poisson evaluation

6.1 An exact product becomes a smooth local interaction

For u near zero define

$$E_\theta(u) = \begin{cases} ((1 + \theta u) \log(1 + \theta u) - \theta u) / \theta, & \theta \neq 0, \\ 0, & \theta = 0, \end{cases} \quad (6.1)$$

and, for v near zero, define

$$b(v) = (1 - v) \log(1 - v) + v, \quad \mathcal{H}(u, v) = E_\theta(u) + sb(v). \quad (6.2)$$

Its derivatives are

$$\begin{aligned} \mathcal{H}_u(u, v) &= \log(1 + \theta u), & \mathcal{H}_v(u, v) &= -s \log(1 - v), \\ \mathcal{H}_{uu}(u, v) &= \frac{\theta}{1 + \theta u}, & \mathcal{H}_{vv}(u, v) &= \frac{s}{1 - v}, & \mathcal{H}_{uv} &= 0. \end{aligned} \quad (6.3) \quad (6.4)$$

The Hessian need not be positive: the tree endpoint has $\theta = -1$. Only boundedness of the Hessian near the origin is used.

Lemma 6.1 (Uniform product approximation). *Uniformly for integer $0 \leq \ell \leq d \leq h = o(n)$,*

$$\prod_{r=1}^{\ell} (n + \theta r) \frac{w_{n-1-d}}{w_{n-1}} = n^{\ell-sd} \exp \{ n \mathcal{H}(\ell/n, d/n) + O(h/n) \}. \quad (6.5)$$

Proof. The exact tail identity gives

$$\frac{w_{n-1-d}}{w_{n-1}} = \prod_{r=1}^d (n + \nu - r)^{-s}.$$

After removing $n^{\ell-sd}$, the logarithm of the left side in (6.5) is

$$\sum_{r=1}^{\ell} \log(1 + \theta r/n) - s \sum_{r=1}^d \log(1 + (\nu - r)/n).$$

The first sum is its corresponding integral plus $O(\ell/n)$. Replacing $1 + (\nu - r)/n$ by $1 - r/n$ changes the second by $O(d/n)$, and its sum-to-integral error is also $O(d/n)$. The integrals are $nE_{\theta}(\ell/n)$ and $nsb(d/n)$. All derivatives involved are uniformly bounded because $h/n \rightarrow 0$. $\square$

Set $x = n^{-s}$. By lemma 6.1,

$$S_{n,J}(h) = (1 + o(1)) \sum_{\substack{m_1, \dots, m_J \geq 0 \\ d = \sum j m_j \leq h}} \prod_{j=1}^J \frac{(na_j x^j)^{m_j}}{m_j!} \exp\{n\mathcal{H}(\ell/n, d/n)\}. \quad (6.6)$$

The relative error is uniform over every summand. This is a finite-dimensional sum, even though the original weight series is divergent.

6.2 A small saddle, defined without global maximization

The implicit function theorem applied to $y - x(1 - B(y))^{-s}$ gives the unique analytic germ in (2.12). For sufficiently small positive x , $y > 0$, $1 - \theta A(y) > 0$, and $1 - B(y) > 0$. Define

$$\beta_j(x) = \frac{a_j y(x)^j}{1 - \theta A(y(x))}, \quad u_*(x) = \sum_{j=1}^J \beta_j(x), \quad v_*(x) = \sum_{j=1}^J j \beta_j(x). \quad (6.7)$$

Then

$$u_* = \frac{A(y)}{1 - \theta A(y)}, \quad v_* = B(y), \quad 1 + \theta u_* = (1 - \theta A(y))^{-1}. \quad (6.8)$$

Equation (2.12) is equivalent to the stationary identities

$$\log \frac{\beta_j}{a_j x^j} = \mathcal{H}_u(u_*, v_*) + j \mathcal{H}_v(u_*, v_*), \quad 1 \leq j \leq J. \quad (6.9)$$

We have $\beta_j = a_j x^j(1 + O(x))$ and $u_*, v_* = O(x)$. No assertion that this stationary point is a global maximum is needed. The next lemma evaluates its neighborhood and bounds the entire remaining cutoff region at the same time.

6.3 A cutoff exponential-integrability lemma

Lemma 6.2 (Poisson tilt remainder). *Fix J . Let $N_1, \dots, N_J$ be independent Poisson variables with means $\mu_j \geq 0$, let*

$$L = \sum_j N_j, \quad D = \sum_j j N_j, \quad v = \sum_j \mu_j,$$

and suppose $v = o(n)$, $h = o(n)$, $h \rightarrow \infty$, and $v/h \rightarrow 0$. Write

$$U = \max(|L - \mathbb{E}L|, |D - \mathbb{E}D|).$$

For every fixed $K_0 > 0$,

$$\mathbb{E}[(\exp(K_0 U^2/n) - 1) \mathbf{1}_{D \leq h}] = o(1), \quad \mathbb{P}(D > h) = o(1). \quad (6.10)$$

Consequently, if a random remainder $\mathcal{E}_n$ on $\{D \leq h\}$ satisfies $|\mathcal{E}_n| \leq K_0 U^2/n$, then

$$\mathbb{E}[\exp(\mathcal{E}_n) \mathbf{1}_{D \leq h}] = 1 + o(1), \quad \mathbb{E}[\exp(\mathcal{E}_n) \mathbf{1}_{D \leq h} - 1] = o(1). \quad (6.11)$$

Proof. For a centered weighted Poisson sum $X = \sum c_j(N_j - \mu_j)$ with $0 \leq c_j \leq J$, its cumulant generating function is

$$\log \mathbb{E} e^{tX} = \sum_j \mu_j (e^{tc_j} - 1 - tc_j).$$

For $|t| \leq 1/(2J)$, Taylor's inequality bounds this by a constant multiple of vt^2 . Applying Chernoff's bound with t proportional to $u/(v+u)$, truncated to that interval, gives constants depending only on J such that

$$\mathbb{P}(U > u) \leq 4 \exp \left\{ -c \frac{u^2}{v+u} \right\} \quad (u > 0). \quad (6.12)$$

When $v = 0$ the variables vanish and the conclusion is immediate; one may otherwise replace v by $v+1$ in this bound.

On $D \leq h$, both $L \leq D$ and $\mathbb{E}D \leq Jv = o(h)$ imply $U \leq 2h$ for all sufficiently large n . Tail integration therefore bounds the first expectation in (6.10) by

$$\int_0^{2h} \frac{2K_0 u}{n} e^{K_0 u^2/n} \mathbb{P}(U > u) du. \quad (6.13)$$

Because $(v+1+h)/n \rightarrow 0$, the positive exponent in this integral is at most half the magnitude of the negative exponent in (6.12), uniformly for $0 \leq u \leq 2h$. On $u \leq v+1$ the resulting integral is $O((v+1)/n)$ by a Gaussian integral. On $u > v+1$ it is $O(1/n)$ after using an exponential tail $\exp(-c'u)$. Both tend to zero.

Markov's inequality gives $\mathbb{P}(D > h) \leq Jv/h = o(1)$. Finally, $|e^z - 1| \leq e^{|z|} - 1$ for real z . Combining these facts yields (6.11). $\square$

Remark 6.3 (Why the cutoff is retained). One must not replace the truncated expectation by $\mathbb{E} \exp(cN^2/n)$ for a nontrivial Poisson variable N and $c > 0$; that expectation is infinite. The condition $D \leq h = o(n)$ is part of the proof, not a dispensable convenience. Lemma 6.2 controls the whole required region without making that invalid extension.

6.4 Evaluation by exponential change of measure

Take independent Poisson variables with means

$$\mu_{j,n} = n\beta_j(n^{-s}), \quad 1 \leq j \leq J. \quad (6.14)$$

Their total mean is $O(n^{1-s})$, or tends to zero when $s > 1$. In either case it is $o(h)$ for the cutoff chosen in (4.5).

Use (6.9) to change the factorial weights in (6.6) from means $na_j x^j$ to means $n\beta_j$. If $L = \sum N_j$ and $D = \sum jN_j$, the result is

$$S_{n,J}(h) = (1 + o(1)) \exp\{n\tilde{P}(x)\} \mathbb{E}[\exp(\mathcal{E}_n) \mathbf{1}_{D \leq h}], \quad (6.15)$$

where

$$\tilde{P}(x) = u_* + \mathcal{H}(u_*, v_*) - u_* \mathcal{H}_u(u_*, v_*) - v_* \mathcal{H}_v(u_*, v_*), \quad (6.16)$$

$$\begin{aligned} \mathcal{E}_n = n[& \mathcal{H}(L/n, D/n) - \mathcal{H}(u_*, v_*) \\ & - \mathcal{H}_u(u_*, v_*)(L/n - u_*) - \mathcal{H}_v(u_*, v_*)(D/n - v_*)]. \end{aligned} \quad (6.17)$$

The Hessian in (6.4) is bounded on all line segments involved when $D \leq h$. Taylor's theorem gives

$$|\mathcal{E}_n| \leq \frac{K_0}{n} ((L - nu_*)^2 + (D - nv_*)^2). \quad (6.18)$$

Thus [lemma 6.2](#) makes the expectation in [\(6.15\)](#) equal to $1 + o(1)$.

Substituting [\(6.8\)](#) and [\(6.3\)](#) into [\(6.16\)](#) simplifies it to

$$\tilde{P}(x) = -\theta^{-1} \log(1 - \theta A(y)) + s(\log(1 - B(y)) + B(y))$$

for $\theta \neq 0$, with the stated continuous limit at zero. This is exactly $P(x)$ in [\(2.13\)](#). Together with [lemma 5.1](#), we have proved the multiplicative pressure formula in [theorem 2.3](#).

7 Extracting the pressure coefficients

7.1 The envelope identity

For positive x and positive $\beta = (\beta_1, \dots, \beta_J)$ near the small stationary point, introduce

$$\mathcal{V}(\beta, x) = \sum_{j=1}^J \beta_j \left(1 + \log \frac{a_j x^j}{\beta_j} \right) + \mathcal{H} \left(\sum_j \beta_j, \sum_j j \beta_j \right). \quad (7.1)$$

Its partial derivatives with respect to β_j vanish at [\(6.7\)](#), precisely by [\(6.9\)](#). At that point its value is $P(x)$, by [\(6.16\)](#). Differentiating the composite function therefore gives

$$xP'(x) = \sum_{j=1}^J j \beta_j(x) = B(y(x)). \quad (7.2)$$

This differentiation is performed first for $x > 0$; both sides are analytic at zero and hence the identity also holds as an identity of power series. It uses stationarity, not a global variational principle.

7.2 Lagrange inversion a second time

Apply formal Lagrange inversion to $y = x(1 - B(y))^{-s}$ and the function B . For $k \geq 1$,

$$[x^k]B(y(x)) = \frac{1}{k} [t^{k-1}]B'(t)(1 - B(t))^{-ks}. \quad (7.3)$$

Since $\Psi'_{ks}(u) = (1 - u)^{-ks}$, equations [\(7.2\)](#) and [\(7.3\)](#) imply

$$\begin{aligned} [x^k]P(x) &= \frac{1}{k^2} [t^{k-1}] \frac{d}{dt} \Psi_{ks}(B(t)) \\ &= \frac{1}{k} [t^k] \Psi_{ks}(B(t)). \end{aligned} \quad (7.4)$$

This proves [\(2.9\)](#). A coefficient of degree k uses only $a_1, \dots, a_k$, proving the finite-jet assertion. Finally, analyticity gives

$$nP(n^{-s}) = \sum_{k=1}^J p_k n^{1-ks} + O(n^{1-(J+1)s}).$$

The remainder tends to zero by [\(2.6\)](#). This completes the proofs of [theorems 2.2](#) and [2.3](#).

7.3 Why the uniform version follows

On a compact positive s -interval, choose once and for all

$$\max(1 - s_{\min}, 0) < \kappa < \alpha < 1.$$

The summable majorant in [lemma 4.1](#), the tail constants, and the insertion estimates are uniform under the stated bounds on the finite head and tail. The strict inequality $(J + 1)s_{\min} > 1$ supplies a common positive decay exponent for [\(5.5\)](#). The implicit function theorem supplies a common neighborhood of zero because J is fixed and the finite coefficients range over a compact set. The Hessian bounds and the Poisson tail constants are also uniform. Finally, Taylor's theorem for P has a common remainder constant on a smaller common neighborhood. These observations prove the uniformity statement following [theorem 2.3](#).

8 Explicit coefficients and every critical point

8.1 The first four universal polynomials

Expanding [\(2.9\)](#) gives

$$p_1 = a_1, \tag{8.1}$$

$$p_2 = a_2 + \frac{s + \theta}{2} a_1^2, \tag{8.2}$$

$$p_3 = a_3 + (2s + \theta)a_1a_2 + \left(\frac{s^2}{2} + s\theta + \frac{s}{6} + \frac{\theta^2}{3}\right)a_1^3, \tag{8.3}$$

$$\begin{aligned} p_4 = & a_4 + (3s + \theta)a_1a_3 + \left(2s + \frac{\theta}{2}\right)a_2^2 \\ & + (4s^2 + 5s\theta + s + \theta^2)a_1^2a_2 \\ & + \left(\frac{2s^3}{3} + 2s^2\theta + \frac{s^2}{2} + \frac{3s\theta^2}{2} + \frac{s\theta}{2} + \frac{s}{12} + \frac{\theta^3}{4}\right)a_1^4. \end{aligned} \tag{8.4}$$

These polynomials are valid for the whole interpolation family. In particular, the terms independent of θ in the higher coefficients must not be dropped when passing between its endpoints. The accompanying exact checker verifies these displayed formulas against the coefficient-extraction formula.

For the reversion case $\theta = 1$, they simplify to

$$p_2 = a_2 + \frac{1 + s}{2} a_1^2, \tag{8.5}$$

$$p_3 = a_3 + (1 + 2s)a_1a_2 + \frac{3s^2 + 7s + 2}{6} a_1^3, \tag{8.6}$$

$$\begin{aligned} p_4 = & a_4 + (1 + 3s)a_1a_3 + \left(\frac{1}{2} + 2s\right)a_2^2 + (4s^2 + 6s + 1)a_1^2a_2 \\ & + \left(\frac{2s^3}{3} + \frac{5s^2}{2} + \frac{25s}{12} + \frac{1}{4}\right)a_1^4. \end{aligned} \tag{8.7}$$

8.2 Reciprocal-integer constants

Corollary 8.1 (Closed critical coefficient). *At $s = 1/r$, the coefficient of the constant term n^{1-rs} is*

$$\boxed{p_r = \frac{1}{r} [t^r] \log \frac{1 - \theta A(t)}{1 - \theta A(t) - tA'(t)}}. \tag{8.8}$$

Its coefficient of the monomial a_1^r is

$$[a_1^r]p_r = \frac{(\theta + 1)^r - \theta^r}{r^2}. \quad (8.9)$$

Proof. At $rs = 1$, $\Psi_{rs}(u) = -\log(1 - u)$. Substitution of $B = tA'/(1 - \theta A)$ into (2.9) proves (8.8). To extract the pure a_1^r coefficient, set $A(t) = a_1 t$ and expand

$$\frac{1}{r} (\log(1 - \theta a_1 t) - \log(1 - (\theta + 1)a_1 t)).$$

The coefficient of t^r is (8.9). □

At the reversion endpoint this coefficient is $(2^r - 1)/r^2$. At the tree endpoint it is $(-1)^{r+1}/r^2$. The latter alternating sign does not contradict positivity of $R_n^{(-1)}$: coefficients in the logarithm of a positive asymptotic quantity need not be positive.

8.3 The missing orders one third and one quarter

Example 8.2 (Order one third). Take $\theta = 1$, $w_j = (j!)^{1/3}$, and arbitrary $C > 0$. Then $a_1 = C$, $a_2 = 2^{1/3}C$, and $a_3 = 6^{1/3}C$. Theorem 2.2 gives exactly (1.2). Equivalently,

$$\begin{aligned} \sup_{f \in \mathcal{B}_{1/3, A, C}} |[z^n]f^{\langle -1 \rangle}| &= CA^{n-1}((n-1)!)^{1/3} \\ &\times \exp \left\{ Cn^{2/3} + \left(2^{1/3}C + \frac{2}{3}C^2 \right) n^{1/3} \right\} \\ &\times \exp \left\{ 6^{1/3}C + \frac{5}{3}2^{1/3}C^2 + \frac{7}{9}C^3 \right\} (1 + o(1)). \end{aligned} \quad (8.10)$$

This is a sharp supremum, attained at every degree by the same series.

Example 8.3 (Order one quarter). For $\theta = 1$, $w_j = (j!)^{1/4}$, and $a_j = C(j!)^{1/4}$,

$$\begin{aligned} \log R_n^{(1)} &= a_1 n^{3/4} + \left(a_2 + \frac{5}{8}a_1^2 \right) n^{1/2} \\ &+ \left(a_3 + \frac{3}{2}a_1 a_2 + \frac{21}{32}a_1^3 \right) n^{1/4} \\ &+ a_4 + \frac{7}{4}a_1 a_3 + a_2^2 + \frac{11}{4}a_1^2 a_2 + \frac{15}{16}a_1^4 + o(1). \end{aligned} \quad (8.11)$$

The lower-order exponent terms cannot be recovered from the leading law $\log R_n \sim Cn^{3/4}$. Each is required for a relative-error asymptotic.

8.4 Recovery of the predecessor's regimes

For $1/2 < s < 1$, only $p_1 n^{1-s}$ survives. With $w_1 = 1$,

$$R_n^{(1)} = \exp(Cn^{1-s})(1 + o(1)).$$

At $s = 1/2$, (2.2) gives $a_2 = C\sqrt{\nu + 2}$, so

$$R_n^{(1)} = \exp \left\{ C\sqrt{n} + C\sqrt{\nu + 2} + \frac{3}{4}C^2 \right\} (1 + o(1)). \quad (8.12)$$

These agree with the predecessor [2]; they are consistency checks, not newly claimed special cases. The new finite formula continues below that boundary without introducing a new argument for each interval $1/(r+1) < s \leq 1/r$.

For the alternative coefficient ball $|f_n| \leq B_0 A^{n-1} (n!)^s$, use $\nu = 1$ in (2.2) and set $C = 2^s B_0$. This conversion changes the constants, even though the n th-root Gevrey type is unchanged.

9 Critical windows and finite-head universality

9.1 Every transition on its natural scale

Theorem 9.1 (All reciprocal critical windows). *Fix a positive integer r , $\theta \geq -1$, and a family of admissible weights and positive constants satisfying the uniform hypotheses of [section 2](#) in a neighborhood of $s = 1/r$. Write $p_k(s)$ for the corresponding pressure coefficients, including the s -dependence of the finite weights. For*

$$s_n = \frac{1}{r} + \frac{t}{\log n}, \quad (9.1)$$

uniformly for t in every bounded real interval,

$$\log R_n^{(\theta)}(s_n) - \sum_{k=1}^{r-1} p_k(s_n) n^{1-ks_n} \longrightarrow p_r(1/r) e^{-rt}. \quad (9.2)$$

Thus the ratio obtained by removing the lower-order exponential terms converges to $\exp(p_r(1/r)e^{-rt})$.

Proof. Choose a small fixed neighborhood of $1/r$ on which $(r+1)s > 1$, and use the uniform theorem with $J = r$. After the indicated subtraction, the expression is $p_r(s_n) n^{1-rs_n} + o(1)$. The power is exactly e^{-rt} , and continuity gives $p_r(s_n) \rightarrow p_r(1/r)$ uniformly for bounded t . $\square$

Remark 9.2 (The lower-order coefficients must move). In (9.2), replacing $p_k(s_n)$ by $p_k(1/r)$ for $k < r$ is generally wrong. An $O(1/\log n)$ coefficient error multiplied by $n^{1-k/r}$ need not vanish. The formula retains the actual moving coefficients and therefore resolves the constant-scale transition without hiding a larger error.

At $r = 1$ and $w_1 = 1$, (9.2) becomes $R_n \rightarrow \exp(Ce^{-t})$. At $r = 2$, after removing Cn^{1-s_n} , the reversion limit is $\exp((C\sqrt{\nu+2} + 3C^2/4)e^{-2t})$. For every larger r , (8.8) computes the corresponding constant by a finite polynomial calculation.

9.2 The precise number of weights that survive

Theorem 9.3 (Finite-head universality). *Fix $s > 0$, $C > 0$, and $\theta \geq -1$. Let w and $\tilde{w}$ be two admissible weight sequences with the same exponent s , but possibly different tail scales, shifts, and finite heads. Set $K = \lfloor 1/s \rfloor$. If*

$$w_j = \tilde{w}_j \quad (1 \leq j \leq K), \quad (9.3)$$

then their individually normalized inverse coefficients satisfy

$$\frac{R_n^{(\theta)}(w)}{R_n^{(\theta)}(\tilde{w})} \longrightarrow 1. \quad (9.4)$$

For $s > 1$, the condition is empty and the conclusion still holds.

Proof. For $s \leq 1$, apply [theorem 2.2](#) with $J = K$. Every surviving coefficient is a polynomial in the same initial weights, so the two logarithms differ by $o(1)$. Exponentiation gives (9.4). For $s > 1$, both normalized sequences tend to 1. $\square$

The individual normalization is important. The theorem compares $[z^n]G_\theta/(Cw_{n-1})$ with its counterpart for $\tilde{w}$; it does not claim equality of the unnormalized inverse coefficients. Changing the factorial shift can produce a power of n in the ratio of the baseline weights.

Proposition 9.4 (Sharp sensitivity to the first changed weight). *Suppose $w_j = \tilde{w}_j$ for $j < r$ and $w_r \neq \tilde{w}_r$. Let $\Delta = C(w_r - \tilde{w}_r)$. If $rs < 1$, then*

$$\log \frac{R_n^{(\theta)}(w)}{R_n^{(\theta)}(\tilde{w})} = \Delta n^{1-rs} + o(n^{1-rs}). \quad (9.5)$$

If $rs = 1$, that ratio tends to e^Δ . If $r > 1/s$, it tends to 1.

Proof. Formula (2.9) has the triangular form

$$p_k = a_k + \text{a polynomial in } a_1, \dots, a_{k-1}. \quad (9.6)$$

Indeed, the $a_k t^k$ term can enter only through the linear term of Ψ and the leading denominator term in B , giving $ka_k/k = a_k$. Thus the first pressure difference is Δ at index r . Later powers of n are strictly smaller. Apply the main theorem and distinguish the three cases. $\square$

Example 9.5 (Tail changes invisible after normalization). For a fixed $s = 1/3$, keep w_1, w_2, w_3 fixed and replace all sufficiently large weights by any $D_* \Gamma(j + \nu + 1)^{1/3}$, with arbitrary positive intervening finite values. The normalized reversion ratio to the original sequence tends to 1. Changing only w_3 by δ instead multiplies the limiting ratio by $\exp(C\delta)$. This separates a genuinely observable head change from an invisible tail change at the chosen normalization.

10 Explicit Poisson centers for the small parts

10.1 The probability model attached to the finite sum

Assign to each ordered composition $(j_1, \dots, j_k)$ of $m = n - 1$ probability proportional to its positive contribution in (3.2). Normalization by $R_n^{(\theta)}$ makes this a probability distribution. When there is a unique part at least $m - h$, remove it and let $M_{j,n}$ count the remaining parts of size j . On the exceptional event, define all these counts to be zero; the choice on that event has no effect on the limits below.

At $\theta = -1$, grouping outdegrees in the Lagrange formula for (2.5) gives exactly the same distribution of the nonzero outdegree counts. Thus the statement can also be read as a theorem about degree counts in a weighted plane tree. One can verify the identification by inserting an independent marking variable for each outdegree into the root equation before applying Lagrange inversion. No assertion about the location of the large vertex is needed.

Theorem 10.1 (An explicit product-Poisson approximation). *Under the hypotheses of theorem 2.2, with J fixed as there, let $Y_{j,n}$ be independent Poisson variables with means*

$$\mu_{j,n} = n \frac{a_j y(n^{-s})^j}{1 - \theta A(y(n^{-s}))}, \quad 1 \leq j \leq J. \quad (10.1)$$

Then

$$d_{\text{TV}} \left(\text{Law}((M_{j,n})_{j=1}^J), \bigotimes_{j=1}^J \text{Pois}(\mu_{j,n}) \right) \rightarrow 0. \quad (10.2)$$

With probability tending to one, every part other than the unique large part has size at most $\lfloor 1/s \rfloor$. If $s > 1$, there are no other parts with probability tending to one.

Proof. Lemmas 4.2 and 5.1 show that the exceptional configurations and those containing a small part of size greater than J have total relative mass $o(1)$. On the remaining multiplicity vectors,

lemma 6.1 changes the unnormalized mass by a uniform factor $1 + o(1)$. The exact Poisson change of measure in (6.15) gives the Radon–Nikodym factor proportional to $\exp(\mathcal{E}_n)\mathbf{1}_{D \leq h}$ relative to the product-Poisson law. By (6.11), this factor tends to 1 in L^1 , and its normalizing constant tends to 1. This proves (10.2).

If $js > 1$, then $\mu_{j,n} \sim a_j n^{1-js} \rightarrow 0$. Together with the finite-size cutoff this removes all sizes greater than $\lfloor 1/s \rfloor$. When $s > 1$ even the first mean tends to zero, so the cloud is empty with probability tending to one. $\square$

10.2 Gaussian and critical Poisson limits

For $js < 1$, the means diverge, and the ordinary Poisson central limit theorem plus (10.2) gives

$$\frac{M_{j,n} - \mu_{j,n}}{\sqrt{\mu_{j,n}}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (10.3)$$

For $js = 1$,

$$M_{j,n} \xrightarrow{d} \text{Pois}(a_j). \quad (10.4)$$

All these limits hold jointly with independent components. These are statements about distributions. Total-variation convergence by itself does not assert convergence of unbounded moments; we call $\mu_{j,n}$ *centering parameters*, without silently identifying them with the exact expectations of $M_{j,n}$.

10.3 All centering corrections by finite arithmetic

Expanding the small solution of (2.12) gives

$$y = x + sa_1 x^2 + O(x^3). \quad (10.5)$$

Consequently

$$\mu_{j,n} = a_j n^{1-js} \left(1 + (\theta + js)a_1 n^{-s} + O(n^{-2s}) \right). \quad (10.6)$$

For small s , the correction in this formula may itself exceed the Gaussian standard deviation, so leading-order centering can fail. The finite implicit equation supplies every further coefficient. More explicitly, if

$$\beta_j(x) = \sum_{k \geq j} b_{j,k} x^k, \quad (10.7)$$

then Lagrange inversion gives

$$b_{j,k} = \frac{1}{k} [t^{k-1}] \frac{d}{dt} \left(\frac{a_j t^j}{1 - \theta A(t)} \right) (1 - B(t))^{-sk}. \quad (10.8)$$

This is again a finite coefficient calculation for every selected k .

Corollary 10.2 (How many centering terms suffice). *Suppose $js < 1$. If an integer $M \geq j$ satisfies*

$$(2M + 2 - j)s > 1, \quad (10.9)$$

then replacing $\mu_{j,n}$ in the numerator of (10.3) by

$$\hat{\mu}_{j,n} = \sum_{k=j}^M b_{j,k} n^{1-ks} \quad (10.10)$$

does not change the limiting standard normal law.

Proof. The omitted tail is $O(n^{1-(M+1)s})$. The standard deviation is asymptotic to $\sqrt{a_j} n^{(1-js)/2}$. Condition (10.9) says exactly that the former order is smaller than the latter. $\square$

The $j = 1, M = 1$ condition is $3s > 1$; the next correction permits $5s > 1$. Thus the hierarchy of increasingly necessary centering terms has its own thresholds. Formula (10.8) replaces separate ad hoc calculations at each such threshold.

11 The weighted-tree specialization

Give every plane rooted tree T weight

$$\text{wt}(T) = \prod_{v \in T} q_{\deg^+(v)}, \quad q_0 = 1, \quad q_j = Cw_j \ (j \geq 1), \quad (11.1)$$

and let Z_n be the sum of weights over trees with n vertices. The standard root decomposition proves formally that $T(z) = \sum_{n \geq 1} Z_n z^n$ satisfies (2.5). Hence $Z_n = [z^n]G_{-1}$, and the main theorem proves

$$Z_n = Cw_{n-1} \exp \left\{ \sum_{k=1}^J p_k(s, -1; \mathbf{a}) n^{1-ks} \right\} (1 + o(1)). \quad (11.2)$$

With $C = 1$ and $w_j = (j!)^s$, the first two exponent terms are

$$n^{1-s} + \left(2^s - \frac{1-s}{2} \right) n^{1-2s}. \quad (11.3)$$

This agrees with the published correction in [5, Remark 2.8]. The explicit formula (2.9) continues it to every required order. For instance, at $s = 1/3$ it yields

$$\begin{aligned} \log \frac{Z_n}{((n-1)!)^{1/3}} &= n^{2/3} + \left(2^{1/3} - \frac{1}{3} \right) n^{1/3} \\ &\quad + 6^{1/3} - \frac{1}{3} 2^{1/3} + \frac{1}{9} + o(1). \end{aligned} \quad (11.4)$$

The corresponding extremal reversion constant in (1.2) has different interaction terms. The interpolation parameter explains both without equating them.

Likewise, (10.6) becomes

$$\mu_{j,n} = j!^s n^{1-js} \left(1 - (1-js)n^{-s} + O(n^{-2s}) \right),$$

consistent with the known first centering correction [5, Theorem 2.6]. The new coefficient formula (10.8) specifies as many further terms as (10.9) requires. The tree conclusions concern degree counts and normalization; additional geometric information about branch shapes or the location of the large vertex is not asserted here.

12 Computation, reproducibility, and interpretation

12.1 A coefficient recurrence without enumerating compositions

For a fixed target degree n , define

$$H_n(z) = (1 - \theta CW(z))^{-n/\theta} = \sum_{k \geq 0} B_{n,k} z^k, \quad (12.1)$$

with the exponential interpretation at $\theta = 0$. Formal differentiation gives $(1 - \theta CW)H'_n = nCW'H_n$. Therefore

$$B_{n,0} = 1, \quad kB_{n,k} = C \sum_{j=1}^k (nj + \theta(k-j))w_j B_{n,k-j}, \quad (12.2)$$

and

$$R_n^{(\theta)} = \frac{B_{n,n-1}}{nCw_{n-1}}. \quad (12.3)$$

For $k \leq n-1$ and $\theta \geq -1$, every multiplier in (12.2) is positive. A single coefficient can be computed in $O(n^2)$ arithmetic operations and $O(n)$ stored numbers. This is an arithmetic-operation count, not a bit-complexity estimate: exact integer and rational entries can become large.

For rational s , θ , and finite a_j , formula (2.9) produces exact rational pressure coefficients whenever the a_j are rational. Factorial weights at nonintegral s need not themselves be rational; the exact tests deliberately use rational finite inputs to check the universal algebra, and do not relabel floating-point factorial powers as exact values.

12.2 What the supplied checks actually verify

The standard-library script `code/verify.py` records 1,066 scalar checks. It verifies both inverse compositions through degree 18 in 12 parameter cases, compares ordered-composition sums with grouped cloud sums, and checks the pressure formula independently through degree 7 in 20 rational cases. It also tests the displayed first four polynomials, the critical pure- a_1 formula in 50 cases through $r = 10$, and all 256 sign patterns in a degree-nine extremal box.

The second script, `code/diagnostics.py`, computes normalized coefficients by a log-scaled version of (12.2). Its 45 recorded rows use $s \in \{1/2, 2/5, 1/3, 1/4, 1/5\}$, $\theta \in \{-1, 0, 1\}$, and $n \in \{100, 1000, 10000\}$, with $C = 1$, $\nu = 0$. Fifteen degree-70 values are independently recomputed at 80 decimal digits. The largest absolute discrepancy between those logarithms and the ordinary floating-point computation was below 6.4×10^{-14} .

These comparisons test implementations and finite identities. They are not interval certificates, a proof of the limits, or evidence that an asymptotic regime has started at a particular finite degree. The analytic proofs, rather than the numerical fit, support the theorems.

12.3 Visible preasymptotic behavior

Let $S_n = \sum_{k=1}^{\lfloor 1/s \rfloor} p_k n^{1-ks}$ for the reversion endpoint. The following table uses $C = 1$, $\nu = 0$. The last column uses the analytic pressure from the same finite polynomial A . A dash means that the diagnostic routine did not locate the small positive real branch at that finite argument; the theorem only guarantees that branch for all sufficiently large n .

The order-one-quarter residual grows sharply between $n = 100$ and $n = 1000$ before decreasing. In particular, it would be misleading to infer the constant term from a low-degree numerical fit. Even at $n = 10000$, a logarithmic error 5.24 corresponds to a large relative coefficient error. The resummed pressure often reduces that error in these examples, but no universal finite-degree superiority theorem is claimed.

12.4 Reproduction and formalization boundary

From the package directory, the exact checks require only Python's standard library:

s	n	$\log R_n$	$\log R_n - S_n$	$\log R_n - nP(n^{-s})$
1/3	100	56.504158	21.322616	—
1/3	1,000	126.387080	2.426436	0.721743
1/3	10,000	511.209609	0.848884	0.239161
1/4	100	83.728129	10.723605	—
1/4	1,000	342.874419	74.579330	56.517991
1/4	10,000	1237.289700	5.243241	0.738160

Table 2: Non-certified logarithmic diagnostics. A residual need not be monotone and need not be small at the displayed degrees.

`python code/verify.py`

The optional diagnostic run additionally requires NumPy, SciPy, and mpmath:

`python code/diagnostics.py --max-n 10000`

The default scripts reproduce the recorded output paths; use their `-output` option for a separate destination. For the exact checker this option is a JSON file path; for diagnostics it is a directory. The manuscript is standalone and does not require generated table files to compile:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

A future proof-assistant implementation has a useful division of labor. The exact Lagrange sum, grouping identity, pressure-coefficient formula, and finite coefficient computations form an algebraic layer. The localization bounds, cutoff insertion lemma, Poisson tail control, and uniform asymptotic statements form a separate analytic layer. The present package supplies no Lean source and makes no kernel-verification claim about either layer.

13 Further research questions

13.1 Effective relative-error bounds

The proofs establish $o(1)$, not a sharp computable onset. Derive an explicit error bound for $R_n^{(\theta)} \exp(-nP(n^{-s})) - 1$ in terms of s , the finite head, and the tail parameters. A useful answer would separate the discarded size tail, the Euler-sum correction, and the quadratic Poisson tilt. The diagnostics show why an effective result would matter in practice.

13.2 The first vanishing corrections

The present pressure gives every nonvanishing term in $\log R_n$. Determine the next terms, including the shift ν , the one-step Euler corrections, and the determinant or cumulant corrections from the Poisson tilt. The natural scales include both powers of n^{-s} and ordinary inverse powers of n , with coincidences at rational s . A uniform calculus should handle these resonances without duplicating terms.

13.3 The joint small-order limit

Here s is fixed away from zero in every uniform statement. When $s = s_n \downarrow 0$, the required number of small sizes diverges. Identify the crossover between finite-dimensional condensation

and a regime with an unbounded collection of active sizes. Simply taking $J = \lfloor 1/s_n \rfloor$ in the present proof is not justified because its constants and analytic neighborhoods depend on J .

13.4 Beyond an exact factorial tail

Replace (2.1) by a local-quotient hypothesis such as $w_{n-1}/w_n = n^{-s}(1 + \varepsilon_n)$. What quantitative control of ε_n over windows of length $O(n^{1-s})$ preserves the same pressure? A pointwise ratio limit alone may hide accumulated corrections. Regularly varying modifications and oscillatory perturbations should be distinguished rather than treated by a single informal heuristic.

13.5 Vanishing head weights and lattice restrictions

Allow some w_j to be zero, with smallest positive index $r_0 > 1$. The first cloud scale then changes from n^{1-s} to n^{1-r_0s} . Determine the exact analogue of the pressure theorem, including congruence restrictions and possible zero inverse coefficients. This would extend the present strictly positive class without assuming that missing sizes can be ignored harmlessly.

13.6 Signs and cancellation beyond the extremum

For a general signed or complex coefficient series, the positive Lagrange model no longer represents the inverse coefficient in absolute value. Find structural noncancellation conditions under which a finite-jet equivalent remains valid. Conversely, classify finite-head changes that can suppress an entire leading exponential scale. The coefficient-ball supremum proved here does not answer this question.

13.7 Several variables and operator coefficients

Extend the fixed-radius, fixed-prefactor extremal problem to formal maps of several variables or to operator-valued coefficients. Type-level stability is weaker than the multiplicative control sought here. A useful theorem would identify which finite tensors replace the scalar head $a_1, \dots, a_K$ and whether a finite-dimensional pressure still captures all surviving subexponential terms.

13.8 The positivity boundary in the interpolation parameter

For $\theta < -1$, the factors in (3.2) can change sign. Determine whether a modified theorem survives in a restricted parameter range, whether new cancellation transitions arise, and which algebraic parts of (2.9) retain an asymptotic interpretation. Formal coefficient identities alone do not settle that analytic issue.

13.9 Quantitative probability and geometric information

Obtain rates in (10.2), moderate deviations for the total cloud size, and sufficiently strong tail control to compare actual moments with the explicit centering parameters. At the tree endpoint, connect the pressure derivatives to geometric observables such as branch profiles. These questions go beyond the count distribution proved here.

13.10 Certified computation and proof-assistant integration

Develop a certified interval solver for the small branch of (2.12), with a proof that the chosen branch is the analytic germ required by the theorem. Formalize the finite-jet extraction and the cutoff Poisson lemma, then connect them to an independently verified Lagrange inversion library. A complete implementation should report separately algebraic exactness, analytic error bounds, and machine verification.

14 Conclusion

A fixed Gevrey coefficient ball contains more asymptotic information than its coefficient type. The missing information is finite but grows as the order decreases: at order s , the initial $\lfloor 1/s \rfloor$ weights and their interactions determine every nonvanishing exponent term. The formula

$$p_k = \frac{1}{k} [t^k] \Psi_{ks} \left(\frac{tA'(t)}{1 - \theta A(t)} \right)$$

computes those terms directly, and the finite analytic pressure packages them into a single local equation.

This supplies a full multiplicative answer to the predecessor's unresolved $0 < s < 1/2$ extremal-reversion problem, while also giving explicit critical windows and centering formulas in the weighted-tree specialization. The decisive steps are the relative finite-size cutoff and the cutoff-preserving Poisson tilt. Neither requires the divergent generating function to acquire an analytic meaning.

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